1. Using the following change of variables

$$x = \frac{S}{\Lambda/\mu}, \quad y = \frac{I}{\Lambda/\mu}, \quad t = (\alpha+\mu)T$$

show that the planar system can be nondimensionalized to the following form:

$$\frac{dx}{dt} = \rho(1-x) - R_0 xy$$
$$\frac{dy}{dt} = (R_0 x - 1)y$$

where ρ and R_0 are some combination of the original parameters, λ, β, α and μ.

My work:
Take change of variable functions and solve for S, I, and T:

$$S = x\frac{\Lambda}{\mu}$$
$$I = y\frac{\Lambda}{\mu}$$
$$T = \frac{t}{\alpha+\mu}$$

Solve for $\frac{dy}{dt}$ first. Directly substitute new change of variable expressions into $\frac{dI}{dT}$:

$$\frac{d(y\Lambda/\mu)}{d(t/(\alpha+\mu))} = \beta\frac{x\Lambda}{\mu}\frac{y\Lambda}{\mu} - \alpha\frac{y\Lambda}{\mu} - \mu\frac{y\Lambda}{\mu}$$

Multiply each side of the equation by the reciprocal $\frac{\mu}{\Lambda(\alpha+\mu)}$ to get $\frac{dy}{dt}$ alone. This results in:

$$\frac{dy}{dt} = \frac{\beta xy\Lambda}{\mu(\alpha+\mu)} - \frac{\alpha y}{\alpha+\mu} - \frac{\mu y}{\alpha+\mu}$$

Group terms with the same denominator together, and factor out y:

$$\frac{dy}{dt} = \frac{\beta xy\Lambda}{\mu(\alpha+\mu)} - y\left(\frac{\alpha+\mu}{\alpha+\mu}\right)$$

Where $\frac{\alpha+\mu}{\alpha+\mu}$ simplifies to 1. Factoring out y results in:

$$\frac{dy}{dt} = y\left(\frac{\beta x\Lambda}{\mu(\alpha+\mu)} - 1\right)$$

Let $R_0 = \frac{\beta\Lambda}{\mu(\alpha+\mu)}$. This results in:

$$\frac{dy}{dt} = (R_0 x - 1)y$$

Solve for $\frac{dx}{dt}$. Substitute change of variables into $\frac{dS}{DT}$:

$$\frac{d(x\Lambda/\mu)}{d(t/(\alpha+\mu))} = \Lambda - \frac{\beta xy\Lambda^2}{\mu^2} - \frac{x\Lambda}{\mu}$$

Multiply each side of the equation by $\frac{\mu}{\Lambda(\alpha+\mu)}$ to get $\frac{dx}{dt}$ alone. After simplifying, results in:

$$\frac{dx}{dt} = \frac{\mu}{\alpha+\mu} - \frac{\beta xy\Lambda}{\mu(\alpha+\mu)} - \frac{x\mu}{\alpha+\mu}$$

Combine terms with the same denominator:

$$\frac{dx}{dt} = \frac{\mu - x\mu}{\alpha + \mu} - \frac{\beta x y \Lambda}{\mu(\alpha + \mu)}$$

Factor out $\frac{\mu}{\alpha+\mu}$:

$$\frac{dx}{dt} = \frac{\mu}{\alpha + \mu}(1 - x) - \frac{\beta x y \Lambda}{\mu(\alpha + \mu)}$$

Let $\rho = \frac{\mu}{\alpha+\mu}$ and $R_0 = \frac{\beta\Lambda}{\mu(\alpha+\mu)}$. This results in:

$$\frac{dx}{dt} = \rho(1 - x) - R_0 x y$$

2. Find the equilibria and state under what conditions they exist.

My work:
Find values of x and y that satisfy $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$, where values that satisfy this are denoted as x^* and y^*. Find critical points that satisfy $f(x^*, y^*) = 0$. For $\frac{dy}{dt}$, $0 = (R_0 x - 1)y$. Let y=0 and plug into $\frac{dx}{dt} = 0$, which results in $\rho(1 - x) - 0 = 0$. x must equal 1 for this expression to be true. This results in the critical point $(1, 0)$. Additionally, for $\frac{dy}{dt}$ where $0 = (R_0 x - 1)y$, let $x = \frac{1}{R_0}$ and plug into $\frac{dx}{dt} = 0$. This results in:

$$\rho(1 - \frac{1}{R_0}) - R_0 \frac{1}{R_0} y = 0$$

$$y = \rho(1 - \frac{1}{R_0})$$

This creates another critical point $(\frac{1}{R_0}, \rho(1 - \frac{1}{R_0}))$. This critical point exists under the condition $R_0 \neq 0$.

3. On two separate graphs, sketch (by hand) the nullclines for $R_0 \in (0, 1)$ and $R_0 > 1$. Include the direction of the vector field (described by the differential equation) along each nullcline. Also include some sample trajectories that illustrate the long term behavior of solutions in the phase plane.

My work:
X nullclines: $\frac{dx}{dt} = 0$, $\rho(1 - x) - R_0 x y = 0$. Solving for y yields:

$$y = \frac{\rho(1 - x)}{R_0 x}$$

Alternatively solving for x yields $x = \frac{\rho}{R_0 y + \rho}$, which creates the same curve. Y nullclines: $\frac{dy}{dt} = 0$, $(R_0 x - 1)y = 0$, which yields $y = 0$ or $x = \frac{1}{R_0}$. Using test points to find direction of the vector field along the nullclines: I used sample parameters of $\rho = 1, R_0 = 0.5$ for the case where $R_0 \in (0, 1)$. I used the point (0.5,2) along the x-nullcline and substituted the values into $\frac{dy}{dt}$ to determine the field sign, which yielded downward facing vectors. I used the same parameters and generated test points (2,1) along $x = \frac{1}{R_0}$, which yielded leftward-facing vectors when substituted into $\frac{dx}{dt}$. (0.5,0) was generated to test the direction along y=0 left of the intersection point, yielding rightward-facing vectors. (2,0) tested the direction right of the intersection along y=0, yielding leftward-facing vectors.

For the case where $R_0 > 1$, I used the sample parameters $\rho = 1, R_0 = 2$. In this case, there is an additional intersection $(\frac{1}{R_0}, \rho(1 - \frac{1}{R_0}))$ along the x-nullcline. I chose (.25,1.5) and (.75,1.67) along $y = \frac{\rho(1-x)}{R_0 x}$ to simulate behavior to the left and right of the intersection. These values generate downwards-facing vectors to the left and upwards-facing to the right. For the y-nullcline $x = \frac{1}{R_0}$,

I generated sample points (.5,1) and (.5,.25). These showed leftwards-facing vectors above the intersection and rightwards-facing vectors below it.

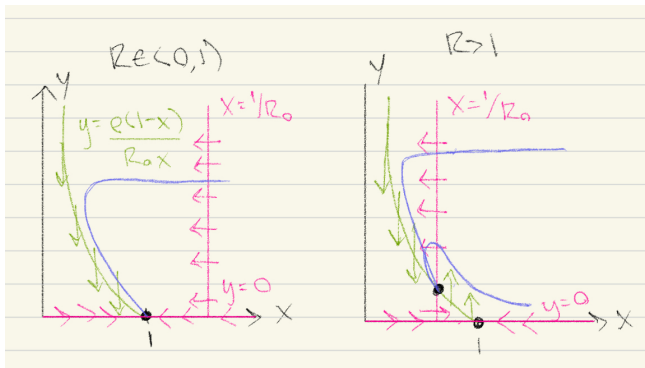

FIGURE 1. Hand drawn nullclines

4. Use the Dulac-Bendixson negative criterion to prove that there are no periodic orbits in the first quadrant of the phase plane.

My work:
Prove ∇Bf does not change signs $\forall x, y > 0$.

$$\nabla Bf = \frac{\partial Bf}{\partial x} + \frac{\partial Bg}{\partial y}$$

where $f(x, y) = \frac{dx}{dt}$ and $g(x, y) = \frac{dy}{dt}$. Allow the smoothing function to be $B = \frac{1}{xy}$ $\forall x, y > 0$. Multiplying by the smoothing function results in:

$$\nabla Bf = \frac{\partial}{\partial x}\left(\frac{1}{xy}(\rho - \rho x - R_0 xy)\right) + \frac{\partial}{\partial y}\left(\frac{1}{xy}(R_0 xy - y)\right)$$

Distributing the smoothing function results in:

$$\nabla Bf = \frac{\partial}{\partial x}\left(\frac{\rho}{xy} - \frac{\rho}{y} - R_0\right) + \frac{\partial}{\partial y}\left(R_0 - \frac{1}{x}\right)$$

Taking the partials results in:

$$\nabla Bf = -\frac{\rho}{x^2 y}$$

Since x and y can only be positive values, ∇Bf will always be negative in the first quadrant.

5. Use PhasePortrait2D to plot the phase plane diagram.

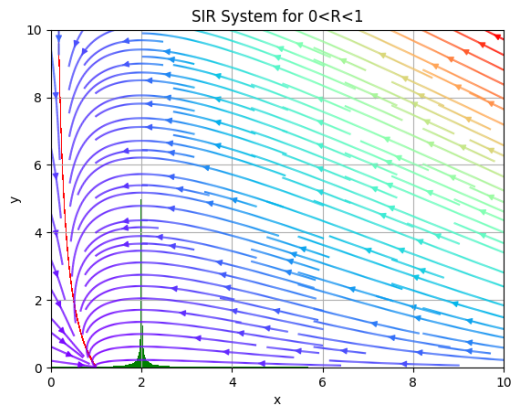

FIGURE 2. Nullclines for $R_0 \in [0, 1]$

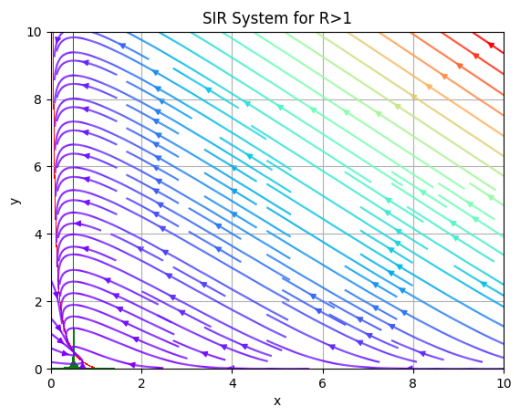

FIGURE 3. Nullclines for $R_0 > 1$

6. Using Python, numerically simulate the full SIR system.

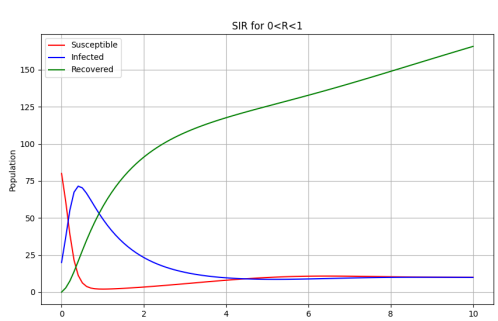

FIGURE 4. Model for $R_0 \in [0, 1]$

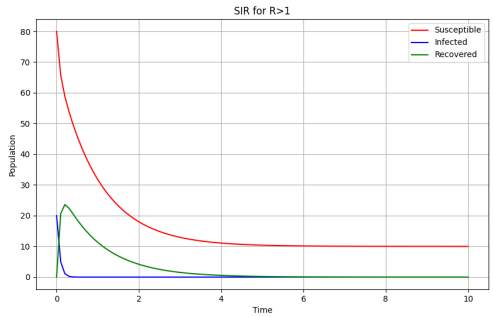

FIGURE 5. Model for $R_0 > 1$

The qualitative behavior for the case where $R_0 \in [0, 1]$ displays the susceptible and the infected populations converging to a constant number of individuals. This means this fixed point represents some individuals who never recover. The recovered population continually increases, which is likely an error in the code. Ideally, this population should also converge to a single steady state. The nullclines for this case show that the fixed points are stable, and so the recovered population should not expand infinitely, as that would represent an unstable node. However, for $R_0 \in [0, 1]$ the nullcline plots have two fixed points, which show the distinct steady populations for infected and susceptible vs. recovered populations.

The qualitative behavior for the case where $R_0 > 1$ shows the infected population as well as the recovered population approach a constant population of 0. The susceptible population approaches the carrying capacity minus the original number of susceptible individuals, effectively losing all of the susceptible population. This plot shows a disease that quickly takes out a whole population. This plot is in agreement with the nullcline plot as the nullclines for $R_0 > 1$ guide vectors towards a singular fixed point.

In conclusion, these two plots show the long-term impacts of R_0 where $R_0 \in [0, 1]$ is a survivable epidemic and $R_0 > 1$ has devastating impacts on the population.

7. Discuss potential alterations to this model.

One limitation is that individuals transfer from susceptible to infected instantaneously. In actuality, there's a period between contracting symptoms and spreading a disease. A discrete delay model could better model the delay between contact and the ability to spread the disease, and would also model a more realistic timeline of disease spread. Another issue is that these differential equations have no terms for a changing population. The population stays constant with the three groups. The model could be improved by adding natural birth and death rates. This would show how a disease spreads with a naturally fluctuating population.